

MARREE, Australia

WMO: 954800

Lat: 29.659S

Lon: 138.068E

Elev: 51

StdP: 100.72

Time Zone: 9.50 (AUA)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.3	3.6	-11.7	1.4	22.7	-9.8	1.6	24.2	12.3	18.6	10.9	17.5	3.2	150	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.6	43.2	20.1	41.6	19.7	39.9	19.3	24.1	30.8	23.0	32.3	22.0	32.9	5.8	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	17.6	25.7	20.7	15.4	25.4	18.8	13.7	25.8	72.8	29.3	68.2	32.1	64.5	32.9	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	11.3	10.1	0.0	45.3	1.1	1.5	-0.8	46.4	-1.4	47.3	-2.0	48.2	-2.7	49.3
			-2.0	25.5	1.2	1.5	-2.9	26.6	-3.5	27.4	-4.2	28.3	-5.0	29.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.4	30.8	29.4	26.1	21.7	16.3	12.6	12.3	14.0	18.1	22.1	25.5	28.1		
	DBStd	7.30	3.96	4.03	3.72	3.45	3.04	2.41	2.59	2.93	3.68	3.93	4.15	4.08		
	HDD10.0	11	0	0	0	0	0	4	6	1	0	0	0	0		
	HDD18.3	653	0	0	1	9	80	174	189	141	49	9	1	0		
	CDD10.0	4154	644	542	498	351	194	81	76	124	242	375	466	561		
	CDD18.3	1754	386	309	240	110	16	1	1	6	42	126	217	302		
	CDH23.3	25558	6122	4608	3155	1262	183	7	22	109	659	1796	3081	4553		
	CDH26.7	15281	4119	2962	1769	537	36	1	1	23	266	909	1788	2872		
(14) Wind	WSAvg	5.0	5.8	5.4	5.1	4.5	4.1	4.0	4.1	4.7	5.2	5.6	5.8	5.9		
(15) Precipitation	PrecAvg	169	25	26	11	11	14	11	12	11	11	13	11	14		
	PrecMax	388	186	203	179	87	52	71	54	61	59	60	46	81		
	PrecMin	39	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	95	38	45	28	20	16	15	14	14	14	16	13	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	44.2	40.3	36.0	29.6	24.3	25.9	29.4	35.0	38.7	42.7	44.2		
		MCWB	20.6	21.2	19.3	17.9	15.6	14.3	14.3	13.9	16.0	17.4	19.0	19.7		
	2%	DB	43.6	42.1	38.4	33.9	27.6	21.8	23.1	26.1	32.3	36.6	39.9	42.1		
		MCWB	20.3	20.5	18.6	17.3	15.3	13.2	12.6	12.9	14.7	16.6	18.2	19.2		
	5%	DB	41.8	40.4	36.5	31.9	25.4	20.1	21.0	23.9	29.8	34.3	37.5	40.0		
		MCWB	20.0	19.7	18.3	16.5	14.2	12.3	11.8	12.3	13.8	15.8	17.5	18.7		
	10%	DB	40.1	38.5	34.7	30.0	23.3	18.6	19.0	21.8	27.2	31.9	35.3	37.8		
		MCWB	19.9	19.6	18.2	15.9	13.4	11.6	11.1	11.4	13.2	15.1	17.0	18.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.8	25.3	24.5	21.5	17.7	16.8	16.7	15.9	18.5	19.9	22.4	25.0		
		MCDB	30.1	29.6	27.6	25.9	24.2	19.2	19.8	23.0	25.2	29.1	31.4	28.8		
	2%	WB	23.9	24.0	22.7	19.8	16.3	15.1	13.8	14.4	16.3	18.2	21.1	22.7		
		MCDB	30.8	32.5	28.8	26.0	23.3	18.6	20.2	23.0	27.2	30.0	30.0	31.2		
	5%	WB	22.7	22.9	21.3	18.3	15.4	13.7	12.4	13.1	15.0	17.1	19.9	21.5		
		MCDB	34.4	33.0	30.4	27.7	22.6	17.7	19.8	22.1	26.9	30.8	30.1	31.8		
	10%	WB	21.7	21.5	19.9	17.0	14.3	12.6	11.5	11.9	13.9	16.1	18.6	20.2		
		MCDB	35.5	33.1	31.4	28.5	21.9	17.3	18.3	20.8	25.5	29.7	31.4	33.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	14.0	13.9	14.1	13.2	12.5	13.7	14.6	15.3	15.4	14.9	14.7		
		MCDBR	16.4	15.5	16.2	16.1	15.1	14.1	16.0	17.4	19.0	18.5	17.4	17.1		
		MCWBR	5.8	5.2	6.2	6.7	7.2	7.7	8.6	8.8	8.6	7.5	6.8	6.2		
	5% WB	MCDBR	12.5	11.7	11.5	12.0	12.4	10.6	13.9	14.8	15.3	14.9	13.4	13.1		
		MCWBR	5.4	4.6	4.9	5.4	6.4	6.3	7.9	8.0	7.3	6.7	5.9	5.6		
(40) Clear-Sky Solar Irradiance	taub		0.366	0.357	0.336	0.325	0.306	0.292	0.297	0.315	0.355	0.356	0.364	0.356		
	taud		2.379	2.435	2.467	2.476	2.496	2.525	2.497	2.429	2.317	2.329	2.349	2.388		
	Ebn at Noon		974	965	954	914	885	877	887	910	918	955	970	987		
	Edh at Noon		130	119	108	97	85	78	84	100	124	131	133	129		
(44) All-Sky Solar Radiation	RadAvg		8.07	7.38	6.45	5.02	3.86	3.31	3.62	4.62	5.91	7.02	7.68	8.09		
	RadStd		0.37	0.49	0.36	0.20	0.20	0.19	0.14	0.18	0.29	0.37	0.36	0.35		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page